Sanjoy Kumar Roy

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EDUCATION

Khulna University of Engineering & Technology (KUET) April 2024
Bachelor of Science in Computer Science and Engineering Khulna, Bangladesh

- **CGPA:** 3.47/4.00
- **Thesis:** *Contextual Language Modeling for Bangla Using Word Embeddings* **(Grade: 4.00/4.00)**

RESEARCH INTERESTS

AI for Software Engineering & Intelligent Systems

- Multimodal fault localization and debugging
- LLM-assisted program comprehension and software maintenance
- On-device AI, semantic retrieval, and intelligent mobile systems

EXPERIENCE

Samsung R&D Institute Bangladesh Ltd. Dhaka, Bangladesh
Senior Software Engineer May 2026 – Present
Software Engineer May 2024 – April 2026

Innovation & Research

- Authored **two invention disclosures** for **SnapSync** (on-device photo sharing) and **TabSense** (browser tab organization) following prior-art research and assessments of novelty, non-obviousness, and business impact; both are under review at Samsung HQ for potential patent filing.
- Developed **four AI proofs of concept (PoCs)** leveraging LLMs, RAG, BERT, and computer vision models (YOLO, MobileFaceNet).
- **ClipSense** (video summarization and contextual search) and **IssueLens** (AI-assisted issue investigation) were selected for PoC exhibition at Samsung's internal AI contest.
- Contribute to team innovation meetings by evaluating peer-submitted ideas and providing constructive technical feedback on novelty, non-obviousness, and implementation feasibility.

Development

- Enhance and maintain Android application-layer functionality **across five core Samsung Internet browser modules** (Settings, Bookmarks, Downloads, History, and Saved Pages), covering UI updates and database maintenance for a browser with **1B+ Google Play downloads**.
- Developed new features and delivered maintenance updates for **three Samsung Internet global market releases (v28–v30)**, supporting **One UI 8, 8.5, and 9**.
- Implemented UI functionality for AI-powered Read Highlights Aloud, result views, and floating UI components across the five modules.
- Resolved **800+ bugs over two years**, addressing memory leaks, crashes, performance bottlenecks, and UX discrepancies to ensure release stability and design compliance.
- Collaborate with engineering teams across India, Brazil, China, Vietnam, and Samsung headquarters in South Korea to coordinate Samsung Internet feature development, integration, and maintenance.

PUBLICATIONS

Contextual Language Modeling for Bangla Using Word Embeddings October 2025

- **S. K. Roy** & K. M. Azharul Hasan, *2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025. DOI: 10.1109/COMPAS67506.2025.11381858.
- Proposed an approach for **contextual word prediction in Bangla** using word embeddings.
- Fine-tuned a pretrained **Bangla Word2Vec model** on domain-specific news text, ranked candidate words using cosine similarity, and compared predictions across **three training corpora of different sizes**.

RESEARCH PROJECTS

Proofs of concept developed collaboratively at Samsung R&D Institute Bangladesh Ltd.

IssueLens: Multimodal AI-Assisted Bug Localization & Investigation | **LLM, RAG** November 2025

- Developed an AI-assisted issue investigation system that analyzes bug descriptions, reproduction videos, and device logs to identify likely responsible modules and suggest probable causes to developers.
- Used FFmpeg to extract timestamped video frames and integrated RAG over source code and feature documentation, enabling the LLM to connect observed failures with relevant implementation details.

SnapSync: Personalized and Simultaneous Photo Sharing System | **Embeddings, P2P** June 2026

- Developed an offline photo-sharing system that groups selected photos by person across nearby devices, enabling previews and simultaneous transfers in one session.
- Integrated face detection and pretrained MobileFaceNet embedding extraction on each phone, with complete-linkage clustering using cosine distance on a coordinator to build shared person groups.

ClipSense: Multimodal Video Summarization & Contextual Search | **LLM, YOLO** March 2025

- Developed a video analysis PoC that automatically generates key-moment highlights and enables contextual search across long videos, returning timestamped segments for direct navigation and playback.
- Engineered an AI pipeline combining LLM-based transcript summarization with YOLO object detection, linking textual and visual evidence to source timestamps for segment selection.

TabSense: BERT-Based Website Classification & Tab Organization | **NLP, BERT** July 2025

- Developed a Samsung Internet proof of concept that used BERT-based topic classification to automatically organize related browser tabs into topic-based groups.
- Collected approximately **10,000 website links**, extracted and categorized page text, and fine-tuned a pretrained BERT model for multiclass topic classification.

PROGRAMMING CONTESTS

- Solved **1,500+ problems** across major competitive programming platforms.
- Participated in **3 onsite** and **110+ online** contests, including an *ICPC Asia Dhaka Regional selection contest*.
- Achieved **4th place (2023)** and **8th place (2022)** as a team member at *KUET Programming Fest*.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, Python, C, C++.
Software Development: Android, UI & Feature Integration, Debugging.
AI & Machine Learning: LLM Applications, NLP, Computer Vision, RAG, PyTorch, OpenCV, YOLOv8.
Data & Tools: MySQL, Pandas, BeautifulSoup, Git, LaTeX.

EXTRACURRICULAR & LEADERSHIP ACTIVITIES

Programming Secretary, SGIPC, KUET May 2023 – April 2024

- Organized programming contests and taught algorithms and data structures to junior programmers.

Vice President, Greater Rangpur-Dinajpur Student Association, KUET May 2023 – April 2024

- Supported regional student welfare and organized admission events and the association's annual tour.